Nassim BENHABIB

ROBOTICS R&D ENGINEER

Advanced Control — Navigation — State Estimation — Autonomous Systems

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Best Paper Award in Automation, ICRA 2020 | First author

PROFILE

Robotics engineer specializing in advanced control, dynamic modeling and autonomous systems, with experience in research at Inria and industrial development. I design and implement control, guidance and motion-planning algorithms in C++, then validate them in simulation and on experimental platforms. My background combines autonomous drone navigation, system identification and physical human–robot assistance through constrained predictive control. I am seeking an R&D role in robotics, control or autonomous systems.

ENGINEERING & RESEARCH EXPERIENCE

UAVIA

Oct. 2022 – Mar. 2026

Autonomous Navigation & Control Engineer

- **Constrained predictive control.** Developed a C++ MPC controller formulated as a quadratic program with qpOASES, accounting for velocity, acceleration, braking and yaw constraints.
- **Motion planning.** Combined global Dijkstra search, local safety corridors and MPC to generate feasible trajectories.
- **Guidance and precision landing.** Designed RTK/IMU-based approach logic with an evolving capture zone, phased descent and contact detection.
- **Safety and degraded modes.** Implemented fallback behaviors and geofencing accounting for restricted areas, wind, position uncertainty and projected fall trajectories.
- **Mission-specific navigation.** Developed terrain-following with altitude adaptation to terrain, and navigation for detecting and covering CO_2 and CH_4 emission plumes.
- **Integration and validation.** Integrated autonomy services with DJI, Parrot, PX4 and ArduPilot; validated them through simulation, telemetry analysis, tuning and flight tests.

Inria Bordeaux

Nov. 2018 – 2022

Doctoral Researcher in Robotics

- **Modeling and optimization.** Built a dynamic model of human–robot–environment interaction; formulated inverse dynamics and force estimation as quadratic programs under contact constraints and physical limits.
- **System identification.** Estimated parameters from measured trajectories and forces; assessed prediction errors on independent experimental trials.
- **Simulation and experimental validation.** Developed a numerical simulator and an instrumented robotic test platform; compared simulated and measured motion and forces under nominal and perturbed conditions.
- **Human–robot assistance control.** Implemented adaptive assistance on the experimental platform using predictive control, quadratic optimization and physical constraints.

ISIR, Sorbonne University

Mar. – Sept. 2018

Robotics Control Intern

- Developed a microrobotic co-manipulation controller and restored an experimental pantograph for validation.

Parrot

May – Sept. 2017

Drone Research Intern

- Prototyped and developed a rapid drone-battery charging system based on an existing setup.

TECHNICAL SKILLS

Control

MPC, quadratic optimization (QP), PID, LQR, adaptive control, guidance.

Modeling and estimation

Dynamics, parameter identification, EKF/UKF, sensor fusion, simulation.

Autonomous robotics

Motion planning, trajectory generation, safety, experimental validation.

Development

C++, Python, MATLAB/Simulink, Linux, ROS, Eigen, Boost, qpOASES, Docker, Gazebo.

Languages

French, English, Arabic.

EDUCATION & PUBLICATIONS

M2 Robotics, Assistance and Mobility
Université Paris-Saclay 2018

M1 Electronics, Energy and Automatic Control
ENS Paris-Saclay 2017

M2 Automatic Control and Industrial Computing
USTO Oran

Doctoral research at Inria Bordeaux, 2018–2022; thesis not yet defended, defense envisaged in early 2027.

Selected publications

Benhabib, Padois, Daney, “Securing Industrial Operators with Collaborative Robots,” IEEE ICRA 2020.

Coauthor, “Task-Consistent Signaling Motions,” HRI 2022.